BOOKS PUBLICATION

- Text Books titled “Introduction to Computer and Information Technology” Alok Publication, 2001.
- Text Books titled ” Programming in C- Alok Publication, 2003
- Text Books titled “Programming in C with Sessional” Alok Publication, 2004
- Text Books titled “Computer Glossary” Orissa Text Book Press, Govt. of Orissa., 2004
- Text Books titled “Management Information System “Himalaya publishing house, New Delhi 2012
- Text Books titled “Information System” Himalaya publishing house, New Delhi for Bharatiya Bidyapith, Distance education program 2012
- Text Books titled “Business Information System”, Himalaya publishing house, New Delhi 2013

Award Received

- Ambedkar Excellence Award 2013 for Eminent Academician on 58th Death anniversary of Dr. B. R Ambedkar at Lohiya Academy, Bhubaneswar
- Orissa Nirman Sanman-2014 on the occasion of Orissa sanskruti aka ananya parampara raja paraba on 15th june 2014
- Mother Teresa Seba Award 2014 on the occasion of 105th birth day of Mother Teresa on 27th Aug 2014
- Orissa Ratna Sanman 2015 on the occasion of workshop on “Importance of Human Value in Social Sector” organized by Print Media
- Sikhyaka Gourava Sanman 2016 On the occasion of “Bijupattnaik Jayanti “Organized by Institute of Engineers, Bhubaneswar on 5th March 2016

Research Guidance

Sl No	**Level of guidance**	**Name**	**Year**	**Status of Degree**
1	PhD.	Virat Chirania	2018	Awarded
2	PhD.	Malaya Malla	2021	Awarded
3	PhD.	Monalisha Patl	2020	Awarded
4	PhD.	Santosh Majhi	2017	Awarded

5	PhD.	Nilayam Kumar Kamila	2017	Awarded
6	PhD.	Prabodh Pradhan	2024	Submitted
7	PhD.	CA Tulika Bal	2021	Awarded
8	PhD.	Rajendra Behera	2021	Awarded
9	PhD.	Gopi Krishna Panda	2022	Awarded
10	PhD.	Santosh Paul	2025	Submitted
11	PhD.	Braja Kishore Mishra	2025	Submitted
12	PhD.	Tushar Kanta Mishra	2026	Submitted

Bibliometric Profile and Systematic Research Analysis of Dr. Sunil Kumar Dhal: A Multi-Decadal Evaluation of Information Technology and Management Science Integration

The professional trajectory of Dr. Sunil Kumar Dhal, currently a Professor within the Faculty of Management Science at Sri Sri University, represents a sophisticated synthesis of mathematical theory, computational engineering, and management intelligence. Spanning over twenty-five years of academic tenure, his career reflects the broader evolution of the Indian information technology sector, moving from the foundational pedagogy of programming languages in the early 2000s to the current frontier of generative artificial intelligence and cloud-based resource optimization. This report provides an exhaustive examination of Dr. Dhal's scholarly contributions, bibliometric standing, and the thematic evolution of his research, specifically as recorded under Scopus Author ID 57189045269 and associated academic repositories.

Scientometric and Academic Evaluation Report: The Interdisciplinary Trajectory, Technological Innovations, and Research Impact of Dr. Sunil Kumar Dhal

Executive Overview of Academic and Professional Trajectory

The academic, pedagogical, and investigative portfolio of Dr. Sunil Kumar Dhal represents a highly complex, multi-decade trajectory that deeply intersects with the rapid evolution of information technology, computational mathematics, and enterprise management systems. Currently positioned as a Professor within the Faculty of Management Studies (and broadly aligned with Management Science) at Sri Sri University in Cuttack, Odisha, Dr. Dhal occupies a unique and highly specialized intellectual space. His work sits precisely at the convergence of foundational computer science, enterprise resource planning (ERP) architectures, and the vanguard of artificial intelligence and deep learning. His formalized academic qualifications underpin a robust, quantitative, and algorithmic foundation, comprising an ascending sequence of degrees: a Master of Science (MSc) in Mathematics, a Post Graduate Diploma in Computer Science and Applications (PGDSC), a Master of Technology (M.Tech) in Computer Science, and ultimately culminating in a Doctor of Philosophy (PhD) in Computer Science.

Integrating into the faculty of Sri Sri University as a regular member on June 16, 2014, Dr. Dhal has consolidated an extensive academic and experimental apparatus, notably directing the operations of "Sunil kumar Dhal's Lab," which serves as an incubator for advanced technical research. His overarching career is defined by a 25-year tenure in the educational and research sectors, a period that spans critical technological epochs. This longitudinal evolution moves sequentially from the dissemination of foundational procedural programming paradigms in the early 2000s, through highly specialized mathematical modeling for pharmaceutical micro-encapsulation, and into contemporary, globally relevant research in unsupervised machine learning architectures, bio-signal processing, and conversational AI for enterprise frameworks.

The subsequent comprehensive evaluation deconstructs his publication metrics, dissects his thematic research clusters, analyzes his pedagogical contributions to textbook literature, outlines his intellectual property portfolio, and assesses his broader institutional influence. The analysis reveals a scholar whose highest bibliometric impact has notably materialized in the later stages of his career, a phenomenon driven by highly strategic pivots into globally critical, high-visibility domains such as sustainable biochar augmentation, cloud virtualization security, and neural-network-driven financial anomaly detection systems.

Scientometric Landscape and Bibliometric Dynamics

A rigorous evaluation of a modern academic's influence necessitates a deep examination of their bibliometric footprint and citation dynamics. The citation patterns associated with Dr. Dhal reveal a distinct and structurally significant curve of exponential late-career impact. This specific bibliometric behavior is typically observed when a seasoned researcher successfully leverages decades of foundational methodological expertise to transition into high-velocity, globally critical research domains, or when laboratory infrastructure matures to a state capable of producing high-frequency, complex computational output.

Citation Dynamics, Indexing Metrics, and Institutional Velocity

As of the current academic indexing period culminating in April 2026, the primary repositories of scholarly literature record a highly active and accelerating profile. According to verified academic profiles linked directly to the institutional domain srisriuniversity.edu.in, Dr. Dhal has accumulated an h-index of 11 and an i10-index of 14. Overall citations stand at 506 across specifically tracked platforms, with an overwhelming and significant concentration manifesting in recent years. Alternate databases that track distinct subsets of the academic literature, such as ResearchGate, record a concentrated subset of 52 publications, 301 localized citations, and a massive engagement metric of over 10,382 platform reads. However, institutional university records indicate a much broader underlying volume of output, citing an extensive 90 national and international journal articles, 26 conference papers, and 2 foundational book chapters.

The most critical indicator of research vitality is not the absolute number of publications, but the longitudinal citation trend. The data demonstrates a profound structural acceleration beginning in the year 2021. This recent window accounts for 374 of his total 506 tracked citations, representing nearly 74% of his measurable bibliometric impact occurring within a highly compressed five-to-six-year timeframe.

Year	Annual Citations	Year-over-Year Growth Trend Analysis
2016	9	Baseline academic visibility

2017	37	Initial acceleration phase; likely linked to early cloud security publications
2018	33	Consolidation of early cloud and AI research
2019	21	Fluctuation; typical of periods between major research cycles
2020	25	Steady state; transitioning into high-complexity ERP and AI models
2021	41	Secondary acceleration onset; post-pandemic technological surge
2022	54	Sustained growth driven by biochar and cloud optimization research
2023	77	High-impact escalation phase

2024	79	Peak output correlation with metaheuristic cloud scheduling
2025	96	Maximum recorded annual citation peak
2026	26 (Year-to-date as of April)	Continuing high trajectory; reflecting immediate impact of 2026 AI papers

Table 1: Longitudinal Citation Trend Analysis (2016–April 2026)

This steep, uninterrupted upward curve from 2021 to 2025 directly correlates with his laboratory's decisive pivot into highly applied artificial intelligence, sustainable agricultural studies, and deep learning models for anomaly detection. The surge strongly implies that the research output from this period successfully intercepted broader scientific macro-trends. When an h-index reaches 11 (with 10 of those high-impact papers published since 2021), it demonstrates that the researcher is not merely publishing into a void, but rather that their specific algorithmic solutions and theoretical models are being actively integrated into the reference structures and foundational literature of subsequent global research endeavors. Currently, 0 articles are listed as available and 1 article is listed as not available based on specific open-access funding mandates, indicating a traditional publishing route relying on institutional access frameworks.

Foundational Research in Pharmaceutical Micro-encapsulation (2007–2014)

A highly unusual, yet theoretically significant, early phase of Dr. Dhal’s published research corpus lies strictly within the domain of physical chemistry and pharmaceutical sciences. Specifically, this era focuses on the complex mechanics of

micro-encapsulation, sustained drug delivery systems, and the precise mathematical characterization of microspheres. Between the years 2007 and 2008, he co-authored multiple intensive studies evaluating the micromeritic properties and solvent evaporation methods of drug delivery mechanisms utilizing specialized polymethacrylate polymers.

Collaborating closely with pharmaceutical researchers such as Bijan Gupta, Dr. Dhal contributed to studies involving the rigorous preparation and evaluation of sustained-release microparticles for highly critical active pharmaceutical ingredients, including Ciprofloxacin hydrochloride, Stavudine, and Glipizide. One notably rigorous investigation explored the physical formulation of Glipizide-loaded Eudragit® RS 100 and RL 100 microspheres via an emulsion solvent evaporation methodology. This research required the assessment of drug loading capabilities, microscopic structural integrity, and the application of Fourier transform analytics to ensure the chemical stability of the encapsulated payload.

Another significant contribution from this period examined the direct effect of processing temperature on the physical characteristics of Eudragit RS PO microspheres. The primary objective was to mathematically and physically optimize the parameters required to extend the stomach residence time of Ciprofloxacin hydrochloride using a highly controlled solvent diffusion technique. The study utilized eight different ratios of Eudragit mixtures, analyzing the resulting flow properties, packability, and the rate at which the drug was retained within the microparticles.

For an academic whose primary formalized qualifications are rooted in Mathematics and Computer Science , deep involvement in pharmaceutical micromeritics initially appears anomalous. However, upon deeper analysis, it suggests a role fundamentally centered on experimental design, advanced statistical optimization, and the computational modeling of physical parameters. This is explicitly corroborated by specific references to his technical expertise in "Particle Size" analytics. This early application of mathematical precision to control physical chemistry laid the epistemological groundwork for his later work in computational optimization algorithms. The underlying theoretical logic—controlling and optimizing the release and flow of physical entities over a temporal dimension—is mathematically analogous to controlling the flow of data packets and optimizing computational tasks in distributed network environments, a field he would pivot to heavily in the subsequent decade. This early work proves a foundational capability in applied mathematics that transcends specific physical mediums.

Cloud Computing Infrastructure, Optimization, and Security Architecture (2015–2025)

As global enterprise IT infrastructures fundamentally shifted away from localized, monolithic servers and toward distributed, virtualized cloud architectures, Dr. Dhal's research trajectory adapted in parallel. He shifted his computational focus to address the critical, mathematically dense bottlenecks of virtualization: dynamic load balancing, NP-hard resource scheduling, and the complex management of security contexts across shifting network topologies. This domain represents a massive portion of his middle-to-late career output and has garnered substantial and sustained academic traction.

The Security of Virtualized Environments

In the realm of cloud security, his 2016 paper, "A study on security vulnerability on cloud platforms," published in *Procedia Computer Science*, serves as a cornerstone of his infrastructure research, accumulating a highly respectable 26 citations. This work systematically addressed the inherent risks of multitenancy and distributed data storage. Alongside this, his concurrent studies on authentication frameworks and threat modeling for Virtual Machine (VM) migration in 2015 and 2016 addressed the profound vulnerabilities introduced when live data environments shift dynamically across different physical server racks.

The 2015 work, establishing a "security context migration framework for Virtual Machine migration" (8 citations), and the subsequent 2016 "authentication framework for securing virtual machine migration" (8 citations), established sophisticated cryptographic mechanisms. When a virtual machine moves, it must carry its security protocols, firewalls, and Intrusion Detection System (IDS) configurations seamlessly to avoid a momentary lapse in defense. Dr. Dhal's mathematical frameworks ensured that cryptographic and access-control contexts are seamlessly and securely transferred alongside the payload, minimizing the attack surface during the highly vulnerable migration transit phase.

Algorithmic Resource Optimization and Metaheuristic Load Balancing

Beyond the static frameworks of security, the mathematical optimization of dynamic cloud resources forms a core, highly cited pillar of his laboratory's output. A notable 2022 paper focused on "Improving Throttled Load Balancing Algorithm in Cloud Computing," earning 10 citations by addressing queuing analysis. Throttled load balancing is essential for preventing server overload by strictly managing the number of requests a virtual machine can handle; Dr. Dhal's improvements likely utilized advanced

queuing theory to smooth out traffic spikes and reduce latency without requiring additional physical hardware.

This trajectory culminated in 2024 with a highly cited publication (20 citations in a very short window) in *IEEE Access*, titled "Optimizing task scheduling and resource utilization in cloud environment: a novel approach combining pattern search with artificial rabbit optimization". The utilization of Artificial Rabbit Optimization (ARO)—a novel, highly advanced metaheuristic algorithm inspired by the survival and foraging strategies of rabbits in nature—combined with classical Pattern Search, highlights a deeply sophisticated approach to solving NP-hard scheduling problems in complex cloud infrastructures. By mathematically minimizing the makespan (the total time to complete a set of tasks) and simultaneously reducing energy consumption, this research directly addresses the thermodynamic and economic constraints of modern hyperscale data centers.

Most recently, in 2025, this evolutionary path continued with the development of an "Intelligent VM Allocation Framework for Handling High-Priority Jobs in Cloud Computing: A Dynamic Load Balancing Approach," co-authored with researchers including J. Laha and S. Pattnaik, published in the *Journal of Information Systems Engineering and Management*. This logical progression—from basic load balancing to advanced, biologically-inspired metaheuristic scheduling, and finally to intelligent, priority-aware dynamic allocation—indicates a profound maturation in his laboratory's capacity to handle and optimize massively complex, distributed computational systems.

Advanced Artificial Intelligence and Deep Learning in Anomaly Detection (2015–2026)

The most contemporarily relevant, technically advanced, and rapidly evolving facet of Dr. Dhal’s entire portfolio encompasses Artificial Intelligence (AI) and Machine Learning (ML), with a distinct and highly specialized focus on anomaly and financial fraud detection. This vector of his research demonstrates a clear, chronological evolutionary path from foundational statistical forecasting models to state-of-the-art, unsupervised deep learning architectures.

Early Implementations: Statistical Forecasting and Kernel Selection

Early applications of predictive data analytics in his portfolio utilized classical, mathematically rigorous statistical models. For instance, his 2016 paper, "Time series data prediction of natural gas consumption using ARIMA model," garnered 18 citations. The utilization of Auto-Regressive Integrated Moving Average (ARIMA) models allowed for precise forecasting of scalar variables based on historical lag. This foundational

predictive capability was later expanded and combined with Artificial Neural Network (ANN) fusion forecasting models in 2018 (8 citations), showcasing an early transition from pure statistics to neural architectures for critical infrastructure monitoring. Similarly, he applied complex state-space computational models, including advanced particle filters, to estimate the volatility of call option prices for portfolio risk analysis. Particle filters represent a sophisticated method for estimating the internal states of dynamical systems, proving his capability in advanced financial mathematics.

As the broader field of machine learning evolved, so did the complexity of the algorithms utilized in his research. In 2015, he published a critical "Comparison of Kernel selection for support vector machines using diabetes dataset" (14 citations). Support Vector Machines (SVMs) are powerful classifiers, but their efficacy depends entirely on the mathematical kernel used to map data into higher dimensions; optimizing this for medical datasets demonstrated an early mastery of feature space manipulation.

The Leap to Complex and Metaheuristic AI Models

By 2022, his focus shifted sharply toward financial anomaly detection with the publication "ANFIS for fraud automobile insurance detection system" (11 citations). The use of an Adaptive Neuro-Fuzzy Inference System (ANFIS) is highly significant; it combines the learning capabilities of artificial neural networks with the human-like reasoning of fuzzy logic. This allows the system to evaluate the complex, non-linear, and often ambiguous data inherent in insurance claims, where fraud does not always present as a binary absolute, but rather as a fuzzy probability. During this highly productive period, he also explored feature selection optimization through the paper "MRMR-SSA: a hybrid approach for optimal feature selection" (14 citations in *Evolutionary Intelligence*). By blending Minimum Redundancy Maximum Relevance (MRMR) principles with advanced metaheuristic algorithms, this research optimized the data ingestion pipeline, ensuring AI models process only the most critical variables, thereby reducing computational overhead and increasing accuracy. Other notable work from this era includes extensive applications of Decision Trees, ISSO, and PKRR to improve detection accuracy across various fraud vectors (2024–2025).

The 2026 Paradigm Shift: Unsupervised Learning and Conversational AI

The year 2026 marks the absolute zenith of this AI trajectory with the development of highly sophisticated, enterprise-grade architectures. In February 2026, his laboratory introduced a groundbreaking model: "TCAE: A Temporal Contextual Autoencoder for Unsupervised Financial Fraud Detection". This represents a massive, paradigm-shifting leap in architectural complexity. Autoencoders are specialized neural networks designed to compress (encode) data into a latent space and then reconstruct (decode) it;

anomalies are detected when the network fails to reconstruct a novel, fraudulent input accurately. By adding *temporal contextual awareness* to this architecture, the TCAE model becomes theoretically capable of detecting highly subtle, time-delayed fraudulent patterns hidden within vast streams of financial transactions. Crucially, because it is an *unsupervised* model, it does not rely on heavily labeled, historically biased datasets, allowing it to detect entirely novel fraud vectors that have never been seen before.

Simultaneously, in March 2026, he published on "U-CAI: A Secure Conversational AI Architecture for Natural Language Interaction with Enterprise ERP Systems". This critical research bridges his deep expertise in enterprise software with state-of-the-art Natural Language Processing (NLP). Traditional ERP systems require complex SQL queries or rigid dashboard navigation; the U-CAI framework proposes a highly secure architecture where end-users can query complex, structured ERP databases using unstructured, conversational human language. This represents the bleeding edge of enterprise usability, drastically reducing the friction between human intent and database execution.

Concurrent research during this period also includes the machine learning-based extraction and classification of complex EEG signals using PyEEG modules (2025). Co-authored with R. Satpathy and others, this indicates an ambitious expansion of his AI methodologies into bio-signal processing, brain-computer interfaces, and computational neuroscience, proving that his algorithmic toolkits are fundamentally domain-agnostic.

Enterprise Resource Planning, Organizational Behavior, and Macro-Societal Systems

Given his foundational positioning within the Faculty of Management Science , Dr. Dhal has consistently produced a parallel stream of literature analyzing the socio-technical impact of information systems on organizational behavior, workforce dynamics, and macro-economic frameworks. This research bridges the critical gap between raw computational capability and human-centric enterprise realities.

The Impact of Enterprise Systems on Organizational Reality

His core management research heavily scrutinizes Enterprise Resource Planning (ERP) systems, moving beyond their technical architecture to assess their financial and human impacts. A highly cited 2020 paper, "A meta-analysis of impact of ERP implementation," accumulated 19 citations. The empirical studies evaluated precisely how ERP integration affects the core financial performance of central public sector enterprises, while concurrent research evaluated the specific psychological factors affecting

employee attitudes toward these massive technological implementations in the workplace.

In the realm of organizational behavior and human resource analytics, his 2015 publication, "A study on employee attrition in BPO industries in India" (8 citations) , provided essential data-driven analytics to understand workforce volatility. The Business Process Outsourcing (BPO) sector is critical to India's economy, but suffers from massive churn; Dr. Dhal applied rigorous statistical methodologies to unearth the underlying causes. This was complemented by later research investigating "Electronic Human Resource Management (E-HRM) Tools Implemented In Different Industry in Odisha" (2017, 12 citations), which analyzed localized technological adoption rates and barriers. By 2021, he advanced this domain by assessing the direct impact of cloud computing on the performance of human resources in the IT sector , and subsequently developed predictive data analytics models for evaluating employee productivity and job satisfaction in November 2022.

Macro-Societal Engineering and Pandemic Response

The application of high-level technology to macro-social systems is another deeply recurrent theme in his portfolio. He has published comprehensive reviews on the state of the art in Cyber-Physical Systems and public utility infrastructure in India (2016, 13 citations). Furthermore, he authored a detailed review of advancements in Smart City technologies, emphasizing high-performance computational applications required to manage urban traffic, energy, and data grids (2022). He has also investigated the sociological impact of digital transformation through studies on transformative e-learning frameworks (2023). Notably, during the height of the global pandemic, he demonstrated immense academic agility by quickly pivoting his time-series forecasting expertise to apply ARIMA models to evaluate the COVID-19 situation in India in September 2021 , providing necessary predictive analytics for epidemiological tracking.

Interdisciplinary Triumphs: Biochar and Sustainable Agriculture

The single most cited paper in his entire portfolio, however, represents a radical interdisciplinary departure that synthesizes environmental science with mathematical optimization: "Enrichment of primary macronutrients in biochar for sustainable agriculture: a review," published in the highly prestigious *Critical Reviews in Environmental Science and Technology* (2022). Accumulating a massive 128 citations in a very short timeframe, this paper resulted from collaboration with domain experts such as Lingaraj Sahoo from the Indian Institute of Technology (IIT) Guwahati.

The review systematically evaluated the chemical and structural augmentation of biochar to create highly efficient, carbon-negative, sustainable agricultural fertilizers. Biochar acts as a vital carbon sink, but optimizing its macronutrient retention requires immense data analysis of various chemical matrices. Dr. Dhal's computational expertise likely provided the framework for evaluating these highly complex biological and chemical interactions. This work, coupled with earlier statistical studies on the genetic reproductive performance of indigenous goats in Northern Odisha (2016, 15 citations) and explorations into the scope of AI in the agriculture sector and farm mechanization in Odisha (2024) , highlights a profound, career-long commitment to regional development and agrarian sustainability, proving his ability to leverage data-structuring capabilities to elevate biological sciences.

Pedagogical Architecture: The Evolution of Educational Literature

An integral, yet often overlooked, component of Dr. Dhal’s 25-year academic career is his extensive contribution to foundational pedagogy through comprehensive textbook authoring. The chronological timeline of his published textbooks provides a fascinating historical map of the evolving Information Technology curriculum in India, moving from the strict mechanics of programming syntax to highly abstract, enterprise-level management systems.

Publication Year	Textbook Title	Publisher	Thematic and Pedagogical Significance
2001	Introduction to Computer and Information Technology	Alok Publication	Addressed the early 2000s surge in foundational computer literacy required by the nascent Indian tech boom.

2003	Programming in C	Alok Publication	Formalized core algorithmic logic, procedural programming, and memory management for engineering students.
2004	Programming in C With Sessional	Alok Publication	Expanded pedagogy to include practical, laboratory-based experiential learning and compilation exercises.
2004	Computer Glossary	Orissa Text Book Press	A critical government-backed initiative to standardize complex technological nomenclature regionally.
2012	Management Information System	Himalaya Publishing House	Shifted pedagogical focus from pure code architecture to the management of enterprise data flow.

2012	Information System	Himalaya Publishing House	Tailored specifically for the Bharatiya Bidyapith Distance Education program, emphasizing highly accessible, remote pedagogy.
2013	Business Information System	Himalaya Publishing House	Synthesized IT infrastructure with high-level corporate strategy and business analytics.

Table 2: Chronological Record of Textbook Publications

The decade-long gap between the early, granular programming texts (2001–2004) and the macro-level management texts (2012–2013) perfectly reflects his personal academic pivot, as well as the maturation of the Indian IT sector. In the early 2000s, the primary requirement was the mass production of competent coders capable of writing C syntax. By 2012, the industry had transitioned from providing basic coding services to architecting complex, multinational enterprise solutions. Dr. Dhal’s textbooks evolved synchronously, equipping the next generation of management students with the necessary conceptual frameworks to navigate and lead complex Business Information Systems. Furthermore, the publication of the "Computer Glossary" by the Orissa Text Book Press, explicitly under the Government of Orissa, demonstrates a profound commitment to foundational education at the state level, establishing the standardized technical terminology necessary for the region's technological development.

Intellectual Property, Patent Portfolio, and Technology Commercialization

Beyond pure academic publishing and pedagogical textbook authoring, a critical metric for evaluating modern scientific impact is the translation of theoretical models into legally protected, commercially viable intellectual property and hardware. The

documented records indicate a robust, highly applied, and geographically diverse patent portfolio.

Dr. Dhal has successfully published 4 distinct Indian Patents and has applied for 3 Australian Patents. The utilization of the Australian intellectual property system is highly strategic; it suggests a clear intent to secure patent rights in international jurisdictions known for their stringent and highly valuable technological protections, potentially signaling a long-term intent for international commercialization, licensing, or technology transfer. While specific institutional data from Sri Sri University regarding immediate income generated from technology transfer currently registers at zero , this is typical for deep-tech innovations. It indicates that while the rigorous ideation, prototyping, and legal protection phases are highly active, the portfolio is currently situated in the vital pre-commercial incubation and venture-building stage.

The university ecosystem in which he operates is clearly geared toward this type of commercialization, tracking innovations across various stages of Technology Readiness Levels (TRL) and fostering start-ups such as Publiconn Digital Pvt Ltd (focused on hyperlocal market spaces). Within this highly entrepreneurial environment, Dr. Dhal leverages his laboratory for practical innovation.

A publicly identifiable, highly relevant, and structurally complete patent within his portfolio is the "Biometric Attendance Device With Touch-Free Body Temperature Screening," successfully registered in May 2022 alongside co-inventors Anand Nayyar, Soumya Ranjan Nayak, and Saraju Prasad. This physical invention represents an immediate, applied cyber-physical engineering response to the severe global epidemiological crisis of COVID-19. By seamlessly fusing advanced biometric identification algorithms (likely relying on his background in machine learning and facial or retinal feature extraction) with non-contact thermal sensory hardware, the device solved a massive enterprise bottleneck. It addressed the critical corporate requirement for secure, hygienic, rapid, and fully automated workforce management during a pandemic. The patent completely embodies the practical, real-world synthesis of his extensive theoretical expertise in cloud-connected hardware (Internet of Things), cryptographic security authentication, and electronic human resource management systems. Furthermore, his execution of 4 formal consultancy projects serves to bridge the gap between the university’s theoretical academic capabilities and the immediate, applied problem-solving requirements of the external corporate or governmental sectors.

Institutional Leadership, Editorial Ecosystem, and Collaborative Networks

An academic of Professor Dhal's extreme seniority and 25-year tenure inevitably transitions from being solely an independent, isolated researcher to becoming a central node and facilitator of broader institutional growth. His influence is heavily propagated through his presence on numerous academic editorial boards, a role that firmly positions him as a critical gatekeeper of academic quality and a primary shaper of localized and international scientific discourse.

The formally recorded editorial board memberships include:

1. **Sri Sri Journal of Management and Innovation:** As a senior faculty member of the publishing university, his role here involves directing the internal academic narrative, setting peer-review standards, and fostering interdisciplinary management research within the institution (Print ISSN: 2319-7587).
2. **Srusti Management Review:** A specialized journal explicitly bridging the gap between Management and Information Technology, published by the Srusti Academy of Management (ISSN NO: 0974-4274). This perfectly reflects his dual-domain expertise.
3. **International Journal of Applied Industrial Engineering:** Published by the prestigious global academic publisher IGI Global (ISSN: 2155-4153), indicating significant international editorial recognition of his engineering optimization frameworks.
4. **International Journal of Knowledgebase Organization:** Also published by IGI Global (ISSN No: 2155-6393), a role directly aligning with his decades of expertise in complex database management, big data architectures, and structured cloud infrastructure.

Furthermore, the existence and management of "Sunil kumar Dhal's Lab" implies a highly structured, well-resourced environment specifically dedicated to the cultivation of doctoral candidates and junior researchers. The administration of a named laboratory typically involves the complex acquisition of computational resources, the strategic direction of multi-year research vectors (such as the continual, documented evolution from foundational ARIMA models in 2016 to the highly advanced TCAE neural networks of 2026), and the systematic, rigorous induction of students into the mechanics of academic publishing. His monumental list of national and international journal articles (totaling 90) and conference papers (26) serves as a vast, internal repository of foundational literature for emerging scholars operating under his guidance within the university framework.

The Architecture of Collaboration

The bibliometric impact and technological reach of Dr. Dhal are deeply intertwined with a highly strategic, expansive, and interdisciplinary co-authorship network. Analyzing these collaboration patterns is absolutely essential for understanding the diffusion of his algorithmic knowledge across different scientific fields.

Notable Academic Collaborator	Institutional / Corporate Affiliation	Associated Research Domains and Strategic Benefit
Dr. Santosh Kumar Majhi	Guru Ghasidas Vishwavidyalaya, Bilaspur	Advanced computing architectures, computational optimization methodologies.
Padmalochan Bera	IIT Bhubaneswar	Computer Science, specifically focusing on advanced cloud and network security architectures. Provides access to tier-one engineering perspectives.
Lingaraj Sahoo	IIT Guwahati	Biotechnology, sustainable agriculture, biochar macronutrients. Facilitated the highest-cited paper in the portfolio (128 citations).

Nilayam Kumar Kamila	Capital One (Corporate Sector)	Financial data analytics, ensuring AI models scale to real-world enterprise realities.
Vijay Kumar Sutrakar	Scientific Research / Aviation / LMAeSI	Multi-disciplinary technological applications and high-level systems engineering.
Anand Nayyar & Soumya Ranjan Nayak	Academic/Research	Intellectual Property development, physical Biometric systems, applied AI hardware.
Rabinaarayan Satpathy	Sri Sri University	EEG signal classification, machine learning. Fosters deep internal university collaboration and multidisciplinary lab integration.

Table 3: Key Academic and Corporate Collaborations

This collaborative matrix reveals several distinct strategic advantages that have propelled his recent citation surge. First, collaborations with tier-one national institutions like IIT Bhubaneswar and IIT Guwahati provide crucial access to advanced, multi-disciplinary laboratory infrastructures. These partnerships elevate the methodological rigor of the output, directly resulting in highly cited, universally applicable review papers. Second, co-authorship with corporate practitioners, such as

Nilayam Kumar Kamila from the financial institution Capital One, acts as a vital reality check; it ensures that theoretical machine learning models in financial anomaly detection remain strictly anchored in practical, enterprise-scale realities rather than just theoretical exercises. Finally, deep internal collaboration within Sri Sri University (evidenced by work with R. Satpathy on EEG signals) fosters a highly active, localized research culture, fulfilling his mandate as a senior professor to continuously elevate the institution's overall research profile and metric standing.

Conclusion and Forward-Looking Prognosis

The exhaustive scientometric and analytical review of the documentation surrounding Dr. Sunil Kumar Dhal presents the profile of an academic whose career is defined by extreme agility, continuous pedagogical evolution, and a profound, unifying understanding of mathematical and computational fundamentals. The transition from the statistical optimization of polymethacrylate pharmaceutical microspheres to the architectural design of temporal contextual autoencoders for unsupervised financial fraud detection is vast. Yet, a singular unifying theoretical thread connects these highly disparate fields: the rigorous application of complex, algorithmic logic to control, optimize, and secure the flow of variables. Whether those variables are chemical compounds sustained in the human stomach, petabytes of data packets migrating across vulnerable global cloud servers, or millions of financial transactions flowing through enterprise accounting systems, the underlying requirement for mathematical optimization remains constant.

His bibliometric trajectory is highly notable for its delayed but structurally explosive acceleration. By perfectly aligning his deep mathematical expertise with the massive global demand for research in Artificial Intelligence, Cloud Optimization algorithms (such as Artificial Rabbit Optimization), and Sustainable Agriculture (Biochar enhancements), he has engineered a phenomenal surge in his citation metrics post-2021. The rapid accumulation of 374 citations since 2021 alone suggests that his current research vectors are not just relevant, but deeply integrated into the contemporary scientific zeitgeist.

The portfolio demonstrates a highly effective dual-track impact model. On one track, advanced theoretical and algorithmic innovations are pushed to international, high-impact journals and platforms like IEEE. On the other parallel track, highly localized, deeply applied research is executed to address the specific regional challenges of Odisha—from studying the HR attrition in regional BPO industries to evaluating the electronic management tools in local businesses, and analyzing the genetic reproductive performance of indigenous livestock to support the agrarian economy.

Furthermore, his decisive transition into tangible intellectual property creation, evidenced by 7 combined national and international patent applications , coupled with the actual hardware development of touch-free biometric devices , clearly indicates a final career shift toward cyber-physical implementation and enterprise commercialization. His entrenched presence at Sri Sri University, actively leading his own computational lab and sitting on multiple international editorial boards , cements his role as a foundational pillar in the institution's technological and management research apparatus. As his groundbreaking 2026 publications in Conversational AI architectures and unsupervised deep learning hit the global academic market , the trajectory strongly and unequivocally suggests that his bibliometric impact, technological innovations, and institutional influence will continue to scale exponentially with the global adoption of advanced enterprise artificial intelligence.